

FHV

Vorarlberg University
of Applied Sciences



Laboratory: Electronic Oscillators

REPORT

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1 Abstract

This lab report is about various oscillator circuits. The following chapters describe how the circuit was built and measured and what conclusions can be drawn. The used measuring instruments are described in the appendix. Manufacturer and model can be found there. The oscillators are to be analysed and measured with an oscilloscope. Calculated and measured values are compared after measurement and checked if, and if so, why the calculated values differ from the measured values.

2 Introduction

The measurement series is divided into four individual measurements. The aim of this series is to detect faults at an early stage in the future and thus to take appropriate action to prevent them. The four circuits are composed of:

- Monostable multivibrator with NE555
- Astable multivibrator with NE555
- Astable multivibrator using two inverters
- Quartz oscillator

Among other things, the influence of the measuring device on the circuit, the influence of voltage changes and the exchange of gate components are considered. The components are of the same logic, however, the technology used in the gate is a different one.

4 Monostable multivibrator

4.1 Introduction

The first task of the laboratory was to build a monostable multivibrator on an experimentation board using the NE555 integrated circuit. With this circuit it was possible to measure the pulse duration and the voltage flowing across the capacitor. This measurement should be carried out by using test probes as well as using standard BNC cables. Figure 1 shows how the NE555 is generally structured.

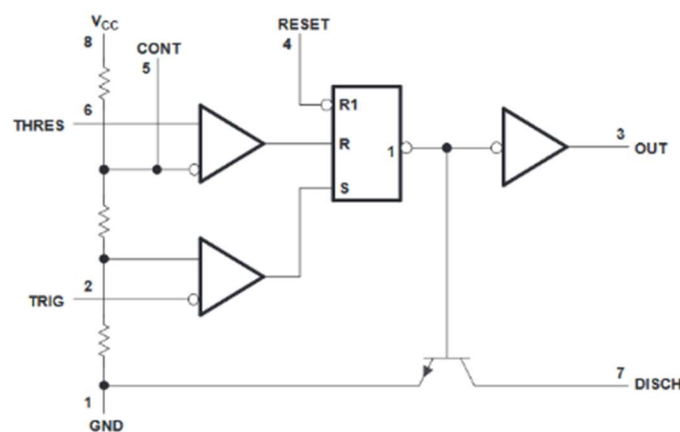


Figure 1 simplified structure of the NE555

It consists of three equal resistors, a flip-flop, two comparators, a bipolar transistor and an inverter. The NE555 timer is a popular and easy to use for general purpose timing applications from $10\mu\text{s}$ to hours or from $<1\text{mHz}$ to 100kHz . In the mono-stable mode of operation, the timed interval is controlled by a single external resistor and capacitor network.

The monostable multivibrator as shown in figure 2 was build.

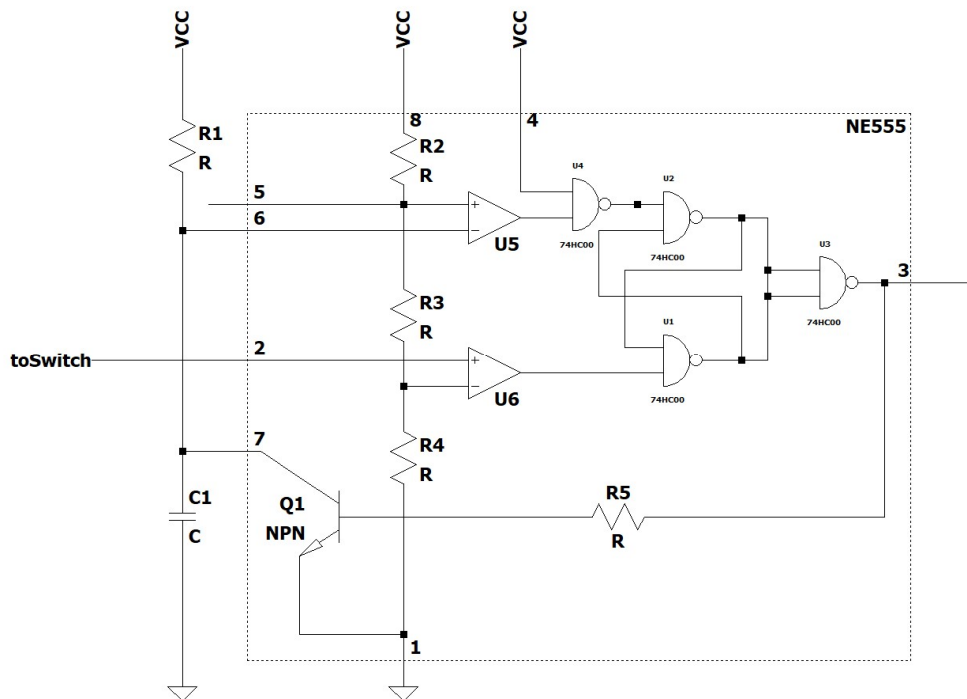


Figure 2 wiring of the NE555

Monostable operation is initiated when TRIG voltage (PIN 2) falls below the trigger threshold. So as soon as the trigger receives a falling edge, the output switches to high until the charging voltage of the capacitor reaches two-thirds of the operating voltage. This characteristic is shown in figure 3.

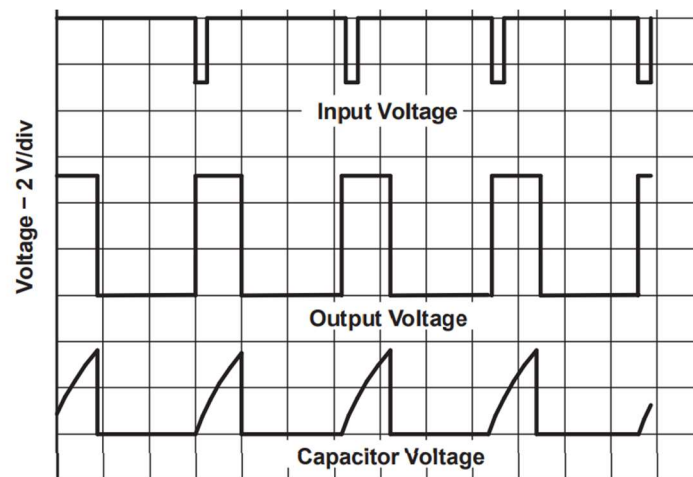


Figure 3 schematic Signal behaviour of the Monostable Multivibrator

The voltage divider at the positive input of the upper comparator K2 results in the voltage level of two thirds of the operating voltage. The transistor T gets a high level as soon as the capacitor reaches

this voltage. Thus, the capacitor is discharged via the ground and the signal at the output becomes low again.

4.2 Procedure

For the circuit a $1M\Omega$ resistor and a $1\mu F$ capacitor were used. 10V was used as voltage supply. So that a falling edge can be generated, a further circuit must be made for the pushbutton as shown in figure 4.

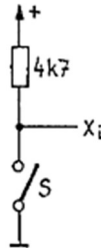


Figure 4 trigger button

To measure the output and the capacitor voltage an oscilloscope was used. The oscilloscope has an input impedance of $1M\Omega$. The probes of the oscilloscope have a 10:1 ratio.

4.3 Results and discussion

When calculating the pulse duration of the output signal, the saturation voltage of the transistor is considered. The performed calculation comes to the following result:

$$U_c(t) = V_{cc} * \left(1 - e^{\frac{t}{RC}}\right)$$

$$t = \ln\left(\frac{V_{cc}}{V_{cc} - U_c(t)}\right) * RC$$

$$t_{0-0,3V} = \ln\left(\frac{10}{10 - 0,3}\right) * 1\mu F * 1M\Omega = 0,03s$$

$$t_{0-6,667V} = \ln\left(\frac{10}{10 - 6,667}\right) * 1\mu F * 1M\Omega = 1,1s$$

$$t_{0,3-6,667V} = t_{0-6,667V} - t_{0-0,3V} = 1,07s$$

The first measurement of the output voltage as well as of the capacitor voltage is made with the help of the test probes of the oscilloscope. The measurement setup is shown in figure 5.

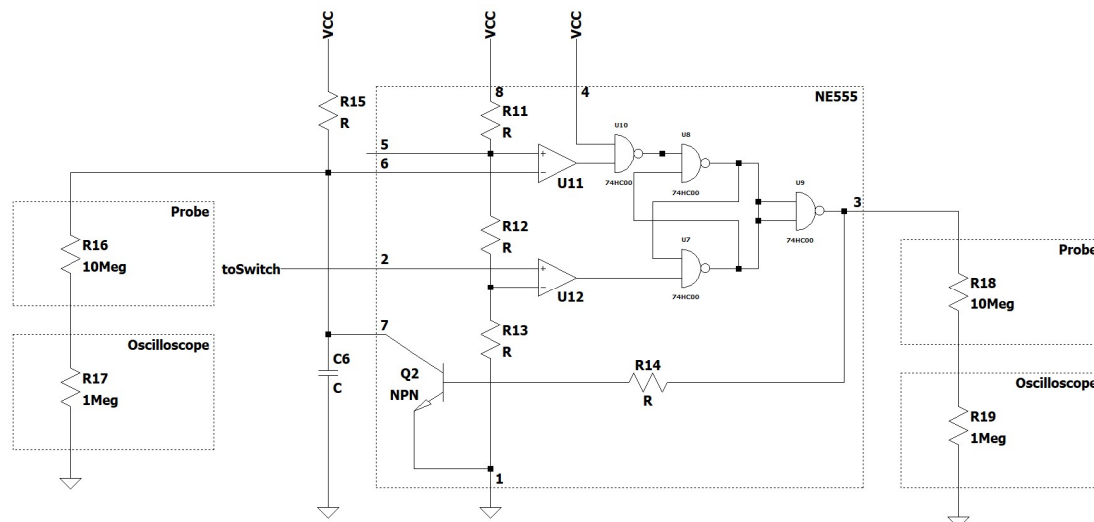


Figure 5 test setup with probes

The result of the pulse duration measurement (1.124s) and the behaviour of the circuit can be seen in figure 6.

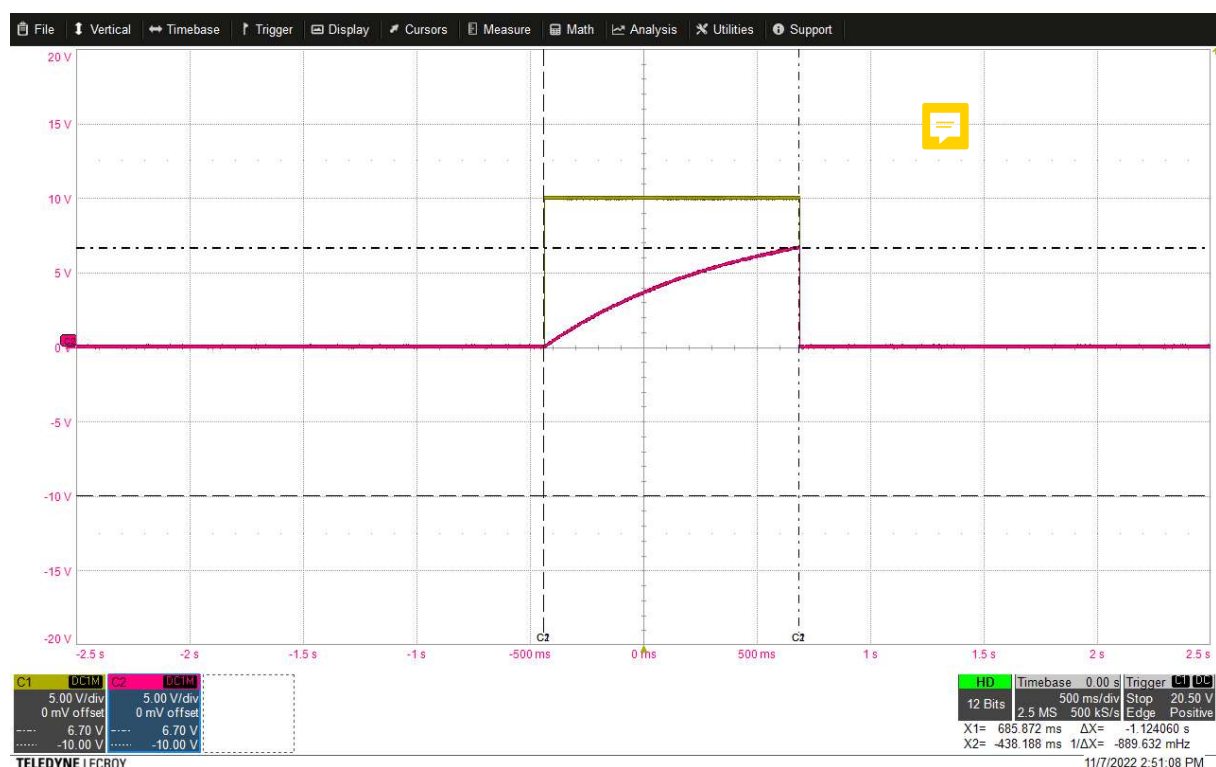


Figure 6 oscilloscope measurement monostable multivibrator

This result does not match with the calculated value. The cause of this deviation depends on several factors:

- Accuracy of the components. Resistors have a very small tolerance ($\pm 1\%$), while capacitors have a higher tolerance ($\pm 20\%$).
- The probes have an impact on the circuit: Because the R15 resistor is 1M the input impedance of the oscilloscope has a big effect on the charging time. The voltage which the capacitor gets is in this case calculated like shown below:

$$U_c = V_{cc} * \frac{10M\Omega}{1M\Omega + 10M\Omega} = 9,09V$$



The low charging voltage results in a longer charging time until the capacitor reaches the voltage of two thirds of the operating voltage.

The second measurement was performed by using the standard BNC cables. This cable can be assumed to be a perfect conductor. That means, the resistance of the cable is 0. The measurement setup is shown in figure 7.

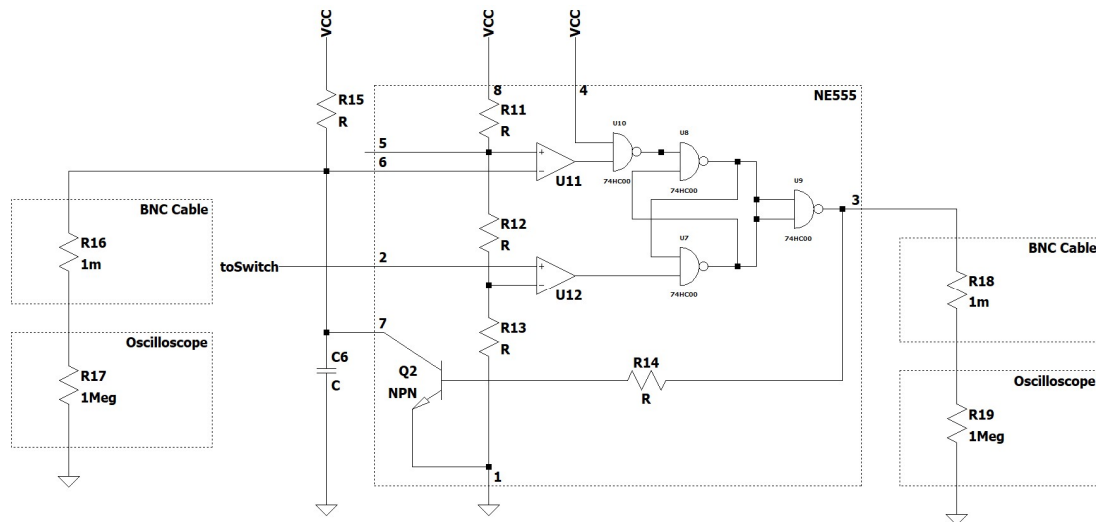


Figure 7 test setup with BNC cable

The result of the measurement with the BNC cable is that the output is always high. The reason is that the capacitor never reaches the two-thirds of the operating voltage. This can be seen in figure 8.



Figure 8 measurement with BNC cables

The reason for this behaviour is that the input impedance of the measuring circuit is $1\text{M}\Omega$. The resistor R15 has the same value. This leads to the problem that the charge voltage for the capacitor drops to 50% of V_{cc} . This means that as described above, the capacitor is never charged to two thirds of V_{cc} . This signifies that the output is always high.



At the third measurement only the output voltage is measured. The result of this measurement is exactly the calculated time of 1.07s. In figure 9 this behaviour is visualized.

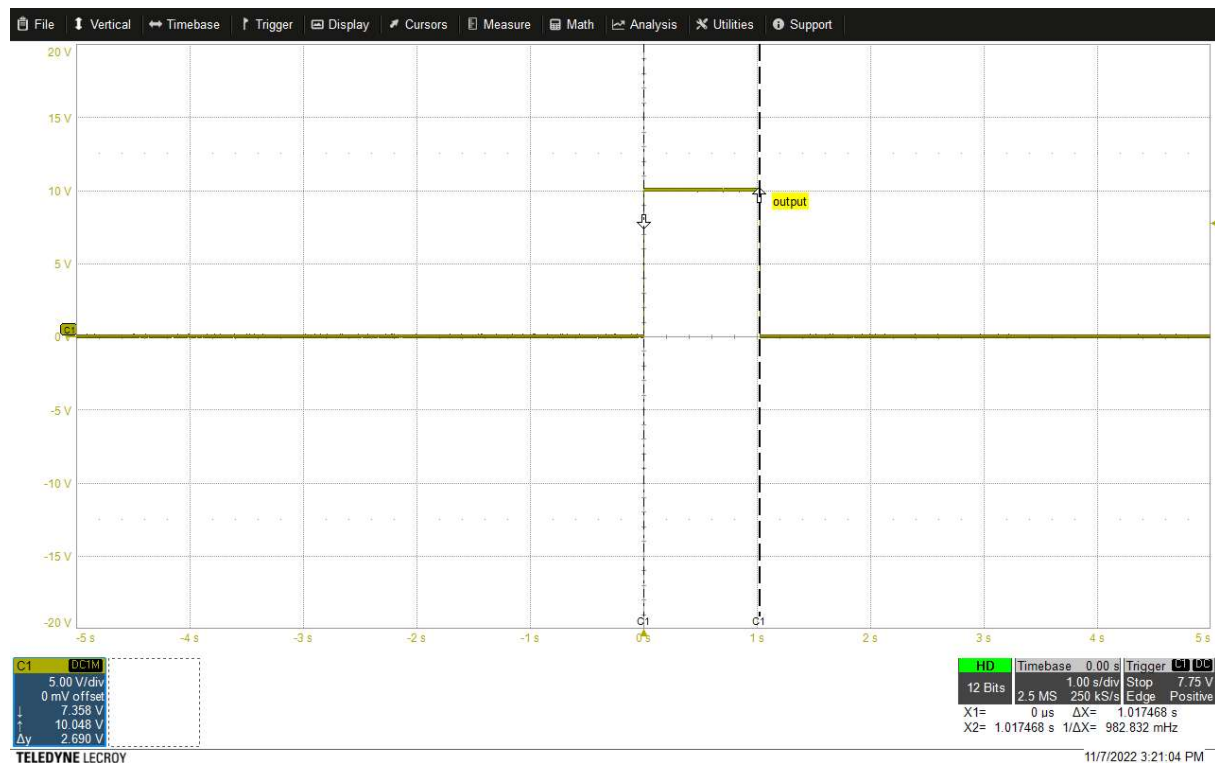


Figure 9 measurement output only

Because the capacitor has no parallel resistance, the charge voltage is now exactly V_{cc} . Thus, the measured result agrees with the calculated result.

At the last measurement of the monostable multivibrator also only the output is measured. The behaviour at different voltage supplies is observed. The voltage supply for this measurement varies between 5V and 15V. When changing the supply voltage, it can be observed that the behaviour of the circuit always remains the same.



4.4 Conclusion

To avoid influencing the measurement setup when measuring the pulse duration, it is best to measure only the output. Otherwise, impedances influence the measuring device and thus the pulse duration.

5 Astable multivibrator with NE555

5.1 Introduction

Given is a circuit as shown in figure 10. The circuit is measured at pin 2 and pin 3. The multivibrator is expected to oscillate at a frequency determined by R1, R2 and C. The following components were used for the measurement:

- R15 = $10\text{k}\Omega$

- $R8 = 10k\Omega$
- $C1 = 100nF$

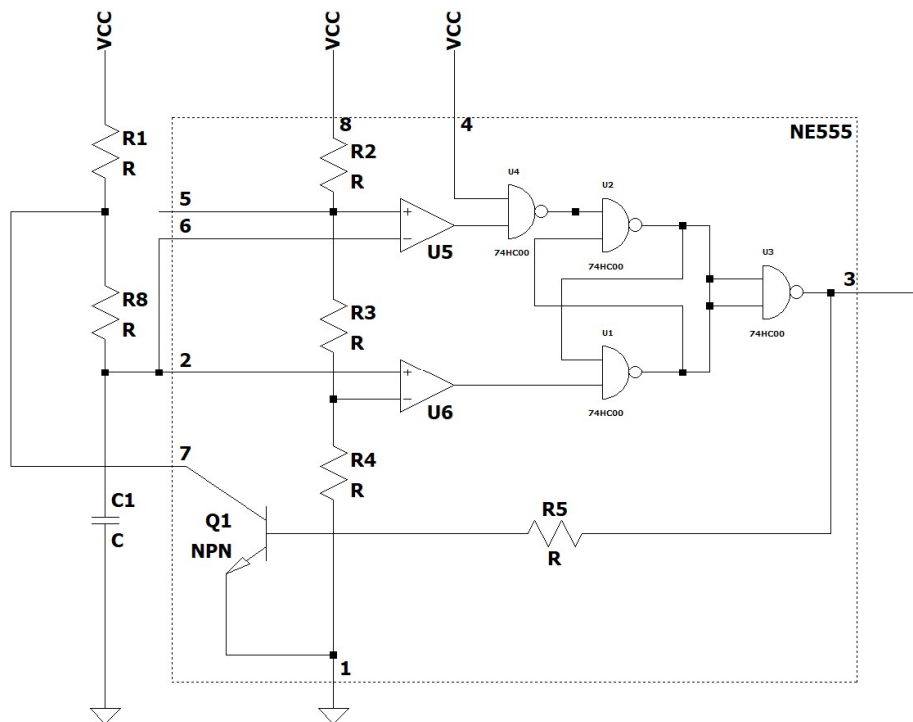


Figure 10 test setup astable multivibrator

5.2 Procedures

The oscillation period can be calculated with the help of equation

$$T = 0,693 * (R1 + 2 * R8) * C1$$

The resulting oscillation period for the given case would be

$$T = 0,693 * (10.000k\Omega + 2 * 10.000k\Omega) * 100nF \approx 2,079ms$$

According to the data sheet, the value calculated in this way is independent of changes in supply voltage.

5.3 Results and discussion

As can be seen in figure 11, the calculated and measured values agree quite well. The absolute deviation of time is

$$\Delta t \approx |2,079ms - 2,089ms| = 0,01ms$$



The percentage deviation in relation to the calculated target value is calculated in

$$\partial t = \frac{100}{2,079ms} * 0,01ms = 2,079\%$$



Figure 11 diagram of astable multivibrator

The supply voltage in figure 11 above was 5V. After changing the voltage to 10V and 15V, there was no change in the oscillation period.

After examining the signal edge, it was found that the voltage only settles after some time ΔT when switching. With a voltage fluctuation of $\Delta U_{\max} = 3.5V$ $\Delta T = 1\mu s$. The problem can be seen in figure 12.

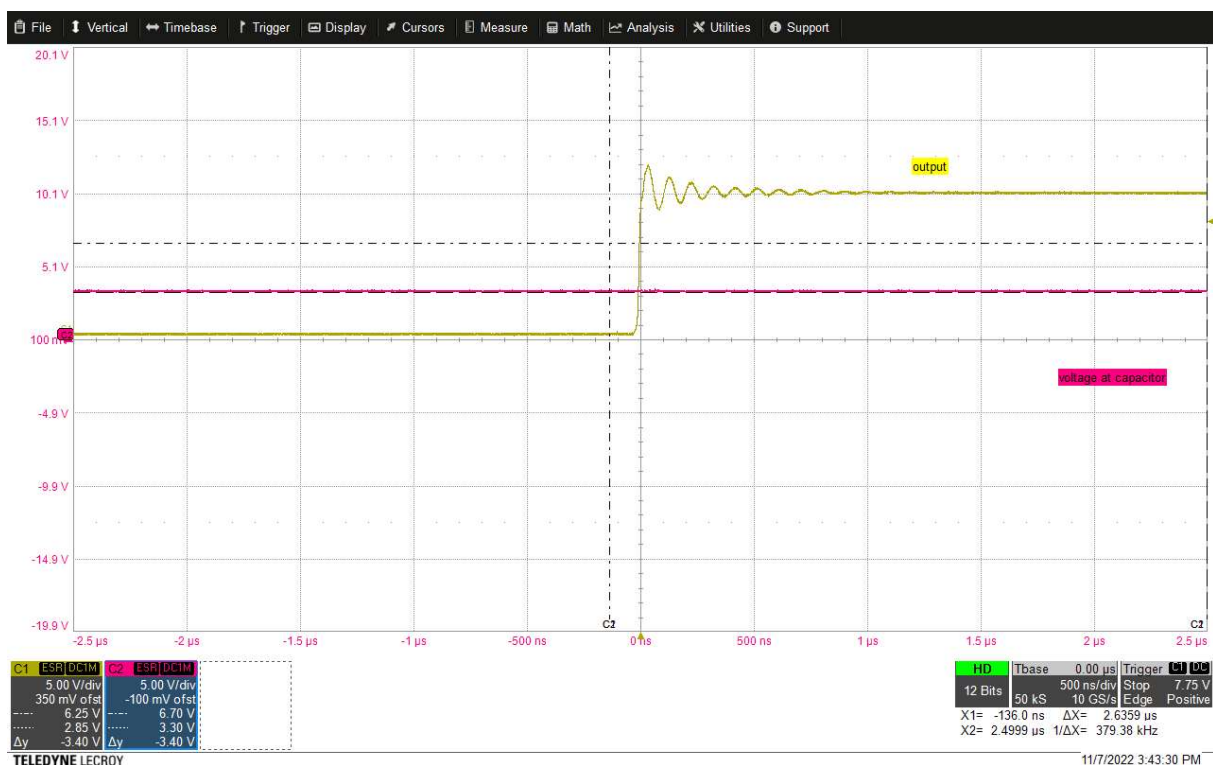


Figure 12 flicker at the edge transitions

To counteract this, a decoupling capacitor between pins 4 and 1 was used to stabilize the supply voltage of the NE555. Capacitors with the capacity of 10nF, 100nF and 1uF were used for this purpose. From the capacity of about 100nF no more improvement could be achieved.

As can be seen in figure 14, the signal is now quite clear and no longer overshooting. This newly gained circuit can be seen in figure 13. The capacitor C2 was added.

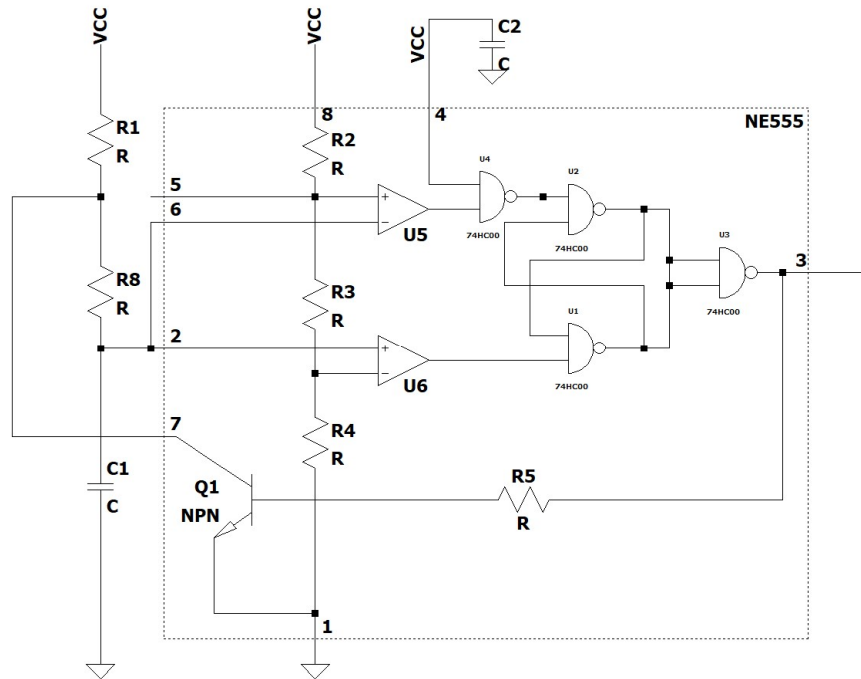


Figure 13 test setup with a decoupling capacitor

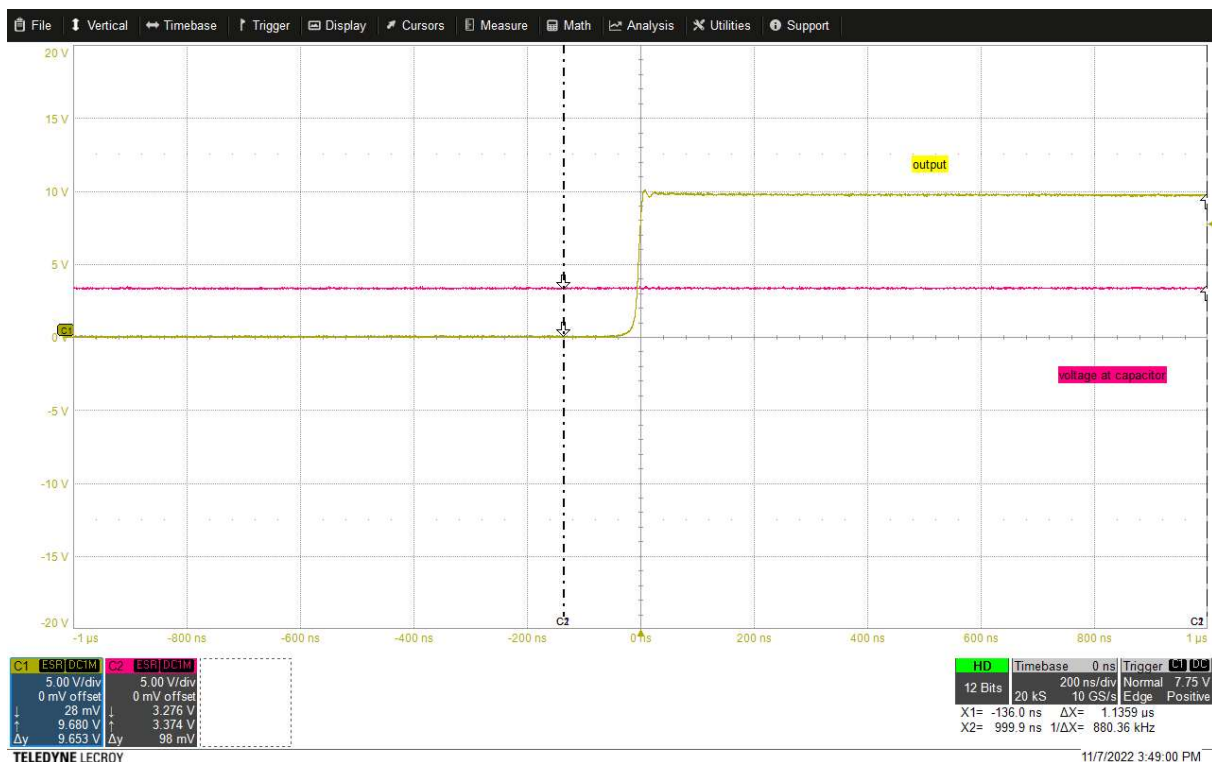


Figure 14 flicker avoided by a supported capacitor

5.4 Conclusion

When using digital modules, care must be taken that very high impulse-like currents flow through the component when switching. This power must be provided by a local capacitor.

6 Astable multivibrator using two inverters

6.1 Introduction

The task of this experiment is to understand the function of the circuit as shown in figure 15. To understand the circuit, the output voltage and the voltage at point P will be measured with the oscilloscope. The voltage curves on point P and S shall be explained. It will also be explained why the resistor R_s is needed.

6.2 Procedure

The function of the following circuit is almost the same as the one described in point 5. It is also a multivibrator. This time, however, two inverters of the type 74C00 are used instead of one NE555. The rest of the components used for the measurement:

- $R6 = 1k\Omega$
- $R7 = 10k\Omega$
- $C3 = 1nF$

The circuit can be seen in figure 15. The supply voltage was at 10 volts.

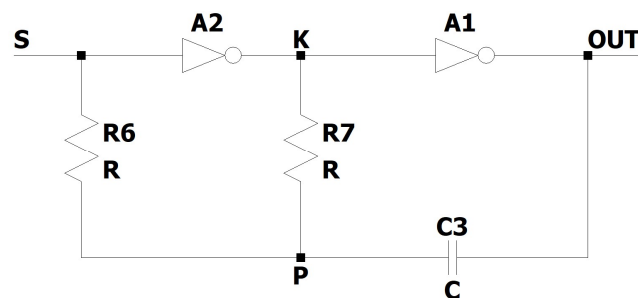


Figure 15 astable multivibrator using two inverters

6.3 Results and discussion

To understand the function of the circuit, the voltages at point S were measured at point P and at the output. Figure 16 shows the voltage over time of the three potentials.

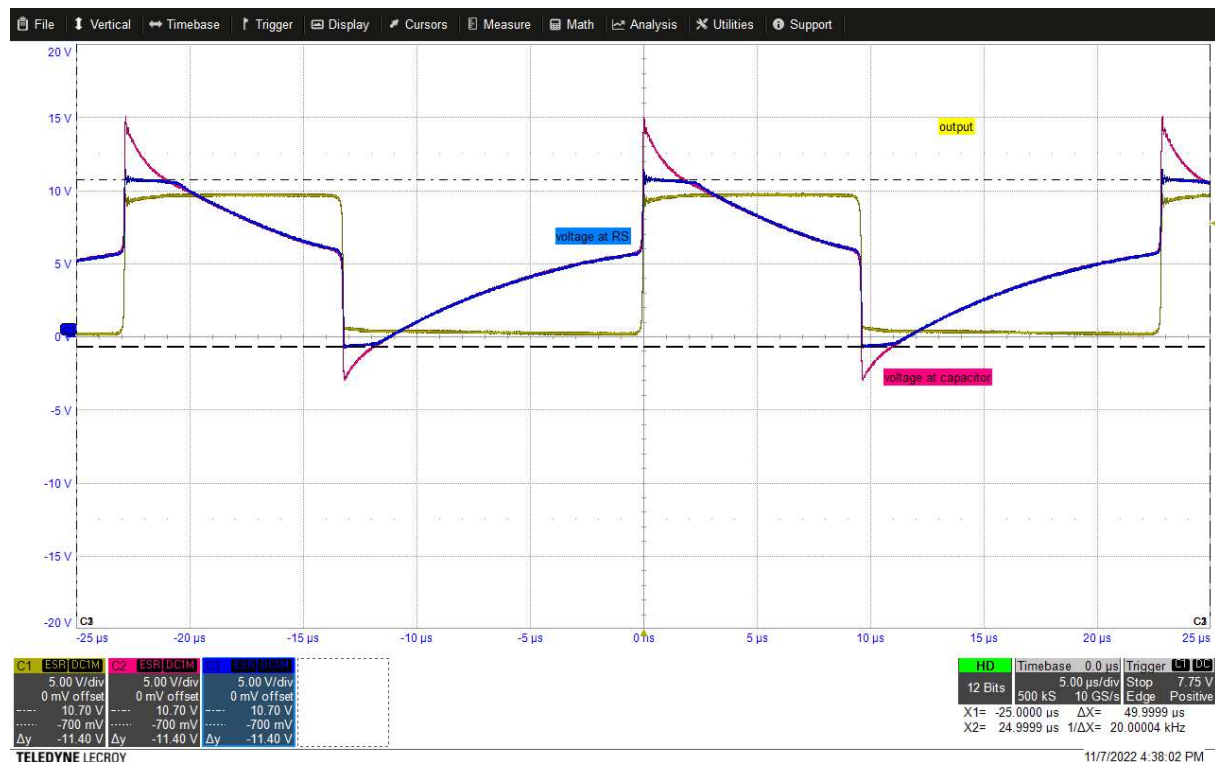


Figure 16 protection circuit for CMOS

In the following interpretation of the process, it is assumed that all components are idealized. The current flowing into the inverter input is therefore set to zero. Figure 16 can be interpreted as follows:

Voltage S and P must be on the same potential because there is no current flow into the inverter. The potential K is therefore inverted to logical high level. Uout is on logical low level. Due to the voltage difference between K and Uout the capacitor is charged with the time constant $R7 * C3$ and the potential P and S increases. If the threshold at the input of the inverter is exceeded, it changes its state. This is reached at about 6V. The potential K is now at logic low level. Uout switches to logic high level. By switching the output, the voltage at the capacitor with respect to the common ground is increased by Vcc. Since the voltage between capacitor and Uout is now $10V + 6V = 16V$, the capacitor discharges through the inverter. Since the voltage at point S is now 16V, this exceeds the allowed maximum voltage for the gate used.

As the voltage is now at 16V, the protection circuit inside the gate becomes low impedance. If the voltage is too high the capacitor gets discharged via Rs. As soon as the voltage has reached a permissible level again, the protection circuit becomes high impedance, and the process starts again. The low impedance of the protection circuit explains the plateau on the oscilloscope graph in detail figure 17.



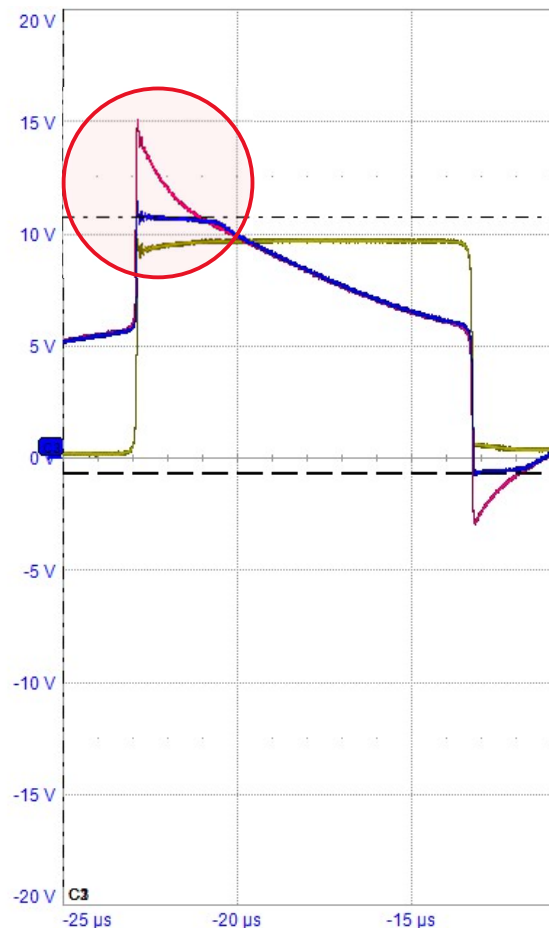


Figure 17 protection circuit

6.4 Conclusion

It is very easy to build an astable multivibrator with two inverters and one resistor. The capacitor stores the charge until the inverters change state. The charge curve thus determines the oscillation time.

7 Quartz oscillator

7.1 Introduction

The next experimental setup was performed with a quartz oscillator. The circuit used can be seen in figure 18. It is expected to measure an oscillating signal at the output. The oscillator used is a 12MHz quartz oscillator.

The two inverters used are 74C00 NAND gates (input shortened) in the first measurement and 74HC00 NAND gates in the second. The rest of the components used for the measurement:

- R9 = 1MΩ
- R10 = 220Ω
- C4 = C2 = 68pF

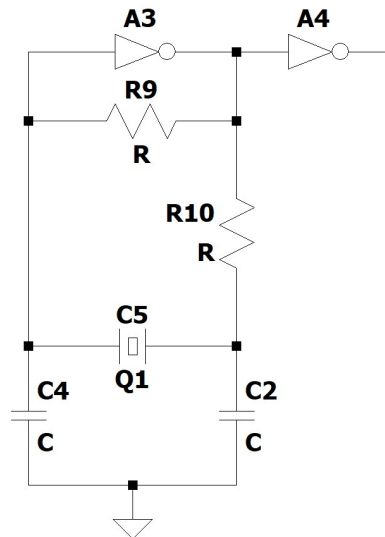


Figure 18 schematic of quartz oscillator

7.2 Procedures

The influence of the supply voltage on the oscillation duration and the edge steepness was checked. Supply voltages of 5V, 10V and 15V were used for this purpose.

7.3 Results and discussion

The result of the measurement with the 74C00 NAND gate can be seen in figure 19.

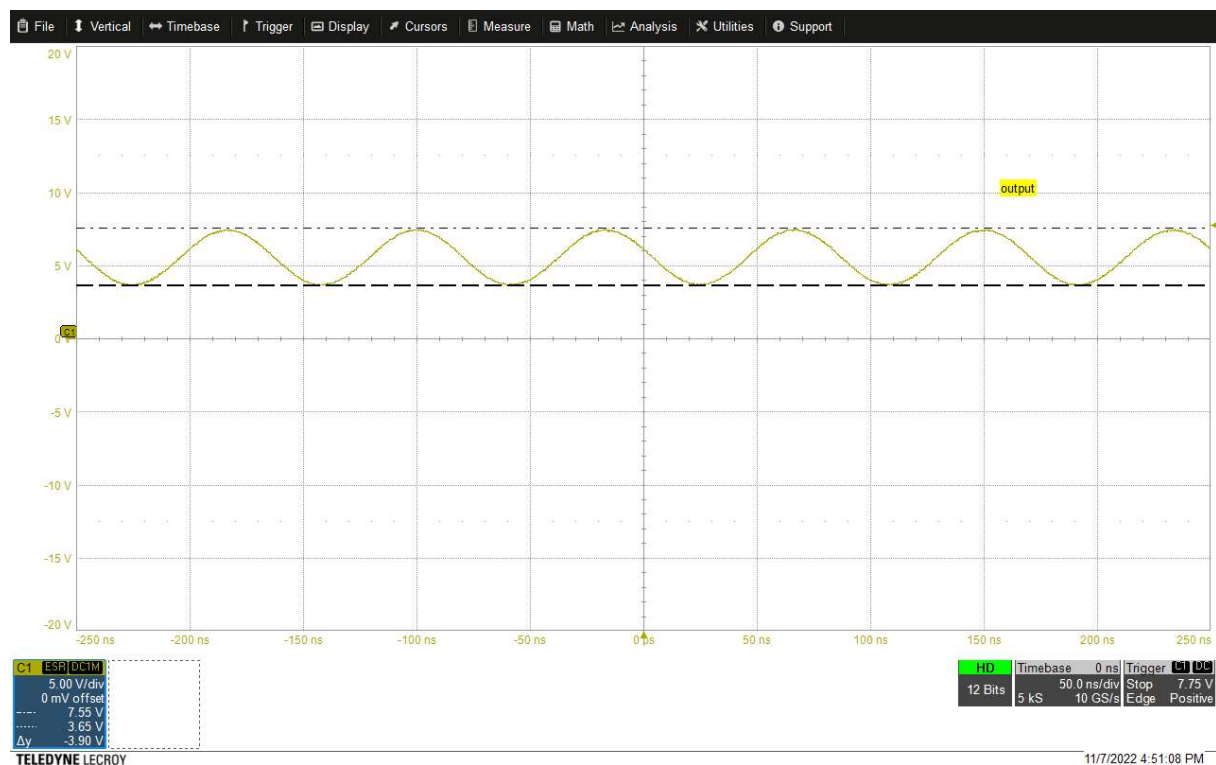


Figure 19 quartz oscillator using the 74C00

The quartz oscillator reacts only at a supply voltage of 15V. If the supply voltage is below this 15V, the quartz does not react at all. A square wave signal is expected, but the result looks more like a sinus signal. The reason for this is the propagation delay time of the 74C00. Typically, the

propagation delay time of a 74C00-circuit is 100ns. But the used crystal needs a period of 83ns. To achieve a faster period time, the faster 74HC00 inverter is used.

For this measurement with the 74HC00 inverter 5V are taken as supply voltage. The resulting diagram can be seen below:

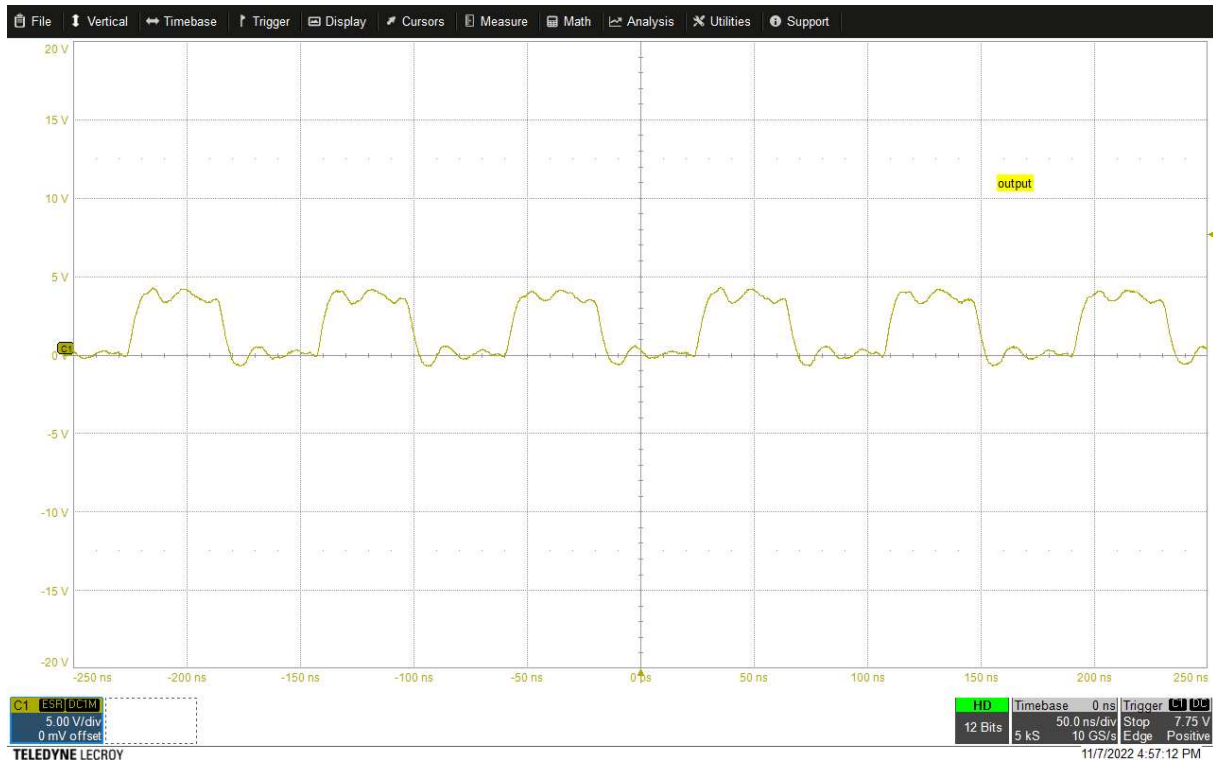


Figure 20 quartz oscillator using 74CH00

A square wave signal was measured. The reason for this is the non-ideal properties of the components.



7.4 Conclusion

When implementing quartz oscillators, the propagation delay time and the power consumption must be considered. The values given by the period times must fit. Otherwise, the oscillator is not going to work properly.

8 References

- Datasheet: NE555 (Texas Instruments)
- Datasheet: 74C00 (Fairchild Semiconductors)
- Datasheet: 74HC00 (Fairchild Semiconductors)
- LT-Spice: self-build schematic

9 Appendices

Laboratory equipment used at working station number 4 in the FH Vorarlberg lab, room U130

- Oscilloscope Teledyne LeCroy HDO6104-MS
- Waveform generator Keysight 33633A
- RLC-Meter Tenma 72-960

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